

ARNAV SINGH

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SUMMARY

Final-year Computer Science student (CGPA: 8.45) specializing in data and business analytics. Built and deployed a fraud detection model (AUC-PR 0.9752, live on Render) and a cohort retention analysis across 540K+ transactions that identified the critical Month 3 churn window. Strong command of SQL, Python, and Power BI, with hands-on experience taking analysis from raw data to production API. Seeking a Data Analyst / Business Analyst internship to drive data-informed decisions at scale.

SKILLS

SQL: CTEs, Window Functions, Cohort Analysis, Subqueries, JOINS, GROUP BY (MySQL, PostgreSQL)

Python: Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn, FastAPI, EDA, Feature Engineering

Visualization: Power BI (DAX, Data Modeling, Interactive Dashboards), Tableau Public, Matplotlib, Seaborn

Statistics: A/B Testing, Hypothesis Testing, Regression, Classification, SMOTE, Model Evaluation

Databases: MySQL, PostgreSQL, Data Pipelines, Star Schema concepts

Tools: GitHub, Jupyter Notebook, VS Code, Excel, Google Sheets, Kaggle

PROJECTS

Credit Card Fraud Detection *Python, XGBoost, SMOTE, SHAP, FastAPI, Render, MLflow* [Live Demo](#); [Project Link](#)

- Built a production-grade fraud detection system on 284K transactions (fraud rate <1%) using XGBoost and SMOTE, achieving AUC-PR of 0.9752, 87.5% recall, and 0.15% false-alarm rate, preventing an estimated £5,338 in fraud losses versus a no-detection baseline.
- Applied SHAP waterfall analysis to explain model decisions at the transaction level, identifying the top 5 behavioral features driving fraud risk, directly actionable for rule-based alert systems.
- Deployed the model as a live REST API (FastAPI + Render) enabling real-time fraud scoring with production-grade inference.

Customer Churn & Cohort Retention Analysis *SQL, Python, Power BI, statsmodels* [Project Link](#)

- Analyzed 540K+ transactions using advanced SQL (CTEs, window functions, cohort analysis) to identify Month 3 as the critical churn window, where 60%+ of at-risk customers dropped off, translating findings into targeted retention strategy recommendations.
- Segmented customers by acquisition cohort and built monthly retention curves in Python (Matplotlib/Seaborn), quantifying revenue risk exposure per cohort and surfacing the highest-impact intervention points.
- Modeled estimated revenue recovery potential from retention improvements, providing a data-backed business case for campaign spend reallocation.

Power BI Sales Performance Dashboard *Power BI, DAX, Data Visualization* [Project Link](#)

- Designed a multi-page interactive Power BI dashboard tracking sales performance across 4 regions and 3 product categories, surfacing discount leakage patterns causing 8-12% estimated margin erosion; recommendations directly adopted by stakeholders.
- Built DAX measures for dynamic KPIs (MoM growth, rolling averages, rank by rep) with drill-through by category, region, and time period, reducing ad-hoc reporting requests through self-serve analysis and a top/bottom SKU revenue waterfall model.

EDUCATION

B.Tech, Computer Engineering - ABES Engineering College, AKTU

2023 - 2027

CGPA: 8.45 | Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Probability & Statistics, Machine Learning

ACHIEVEMENTS & CERTIFICATIONS

- Solved 300+ DSA problems across [LeetCode](#) and [CodeChef](#); CodeChef Rating 1614 (3 Star).
- 2nd Place, ECell Eureka Competition Zonals (IIT Bombay ECell); 3rd Place, Retina AI Hackathon 2026.
- [2025 Google Solution Challenge Participant](#); [Google 5-Day Agentic AI Vibe Coding Contributor](#).
- Certifications:** Master Regression & Prediction with Pandas and Python; Data Science and Machine Learning Fundamentals; SQL & SQL for Data Analytics; Data Fundamentals (IBM SkillsBuild).

LEADERSHIP & COMMUNICATION

Growth & Content Strategist - Aasira (STEM Beyond Labels)

Jul 2024 - Feb 2026

- Grew audience engagement by 35% through targeted social media campaigns; executed 10+ content pieces aligned with brand messaging, tracking performance via weekly dashboards.

Volunteer - GDGoC Member

Oct 2024 - Sep 2025

- Drove a 40% increase in event attendance and 30% improvement in content engagement through data-driven campaign optimization.